Software for Tracking Vulnerable Children

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Abstract* - This paper describes innovation in visualization of a comprehensive software solution (ChildNet) which assists in effective management and tracking of Vulnerable Children in India. This welfare activity particularly focuses on Children in Distress (CID) and their rehabilitation through various initiatives and collaborations with partners/NGOs across India. The main objectives of the System are to help Children in distress and to have complete transparency of the activities by various stakeholders across the programme. This is inline with the Integrated Child Development Scheme (ICDS) which is an integral part of the Ministry of Women and Children Development (Government of India). The innovation was to setup a free national helpline (Tel no. 1098) which would assist in initiating and recording Child Vulnerability Incidents across India. The System is designed to address Child Vulnerability Issues like Abuse & Violence, child Trafficking, Child Labor, child in Conflict with Law, Child Marriage, Child Sexual Abuse, Child without Parental Care, Street Children (without Shelter), , Disabilities, Health issues, Drug Abuse, Gender bias (Girl Child), HIV-AIDS, Missing Children. The System also assists to record collaboration from Government, Public and Private Sectors for addressing this challenge collectively. Comprehensive solution was developed to assist programme in aligning priorities of initiatives and track the progress made in each of the child Intervention case. This resulted in bringing about visibility and accountability with traceability for the initiative with trend analysis. The system is presently operational in 170 cities in 30 States and Union Territories in India and has assisted to address more than 10 million calls so far.

Keywords – ChildNet; Children in Distress; Integrated Child Development Scheme; Child Vulnerability; Child Trafficking; Child labor; Missing Children; Social Justice; Empowerment; United Nations.

I. INTRODUCTION

Tata Consultancy Services Limited (TCS) was approached by Childline India Foundation (CIF) to assist them in articulation of a comprehensive Child Tracking System (ChildNet) which would help them in doing retrospective analysis on societal trends faced by children in distress. The system would also bring visibility and accountability of the programme and their affiliation with their partners.

CIF was founded by a professor from the Tata Institute of Social Science (TISS), Mumbai. CIF was initiated in June 1996 as an experimental project, for the department of Family and Child Welfare of TISS.

In 1998 CIF partnered with the Ministry of Social Justice and Empowerment, Government of India.

CIF primary goal was to respond to emergency needs of every child in need of care and protection throughout the country and for contributing towards protecting the rights of children.

On an average, over 40,000 children in India are reported missing every year, of which approximately 11,000 remain untraced. (Reference: Report on Child Rights, New Delhi 2005). The average at which there is an increase in incidence has risen by at an average 12% yearly. These are stark figures which need to have immediate attention.

CIF has facilitated a 24 hr nation-wide free phone outreach emergency helpline (1098) for children in need of care and protection. Project was later supported by the Union Ministry for Women and Child Development and also linked State Governments, NGOs, Bilateral/Multilateral Agencies and Corporate Sector.

CIF functions through a network of over 415 organisations spread across states of the country. The ChildNet System is operational in 170 cities in 30 states and Union Territories in India and has assisted to address approximately 2 millions calls each year (currently). The total calls where received till date is over than 15 million calls (since 2000). Two types of calls are facilitated – counseling (which entail basic awareness and guidance to child) and intervention (where specific intervention is required for assisting the child).

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II. OBJECTIVE

CIF has undertaken a highly challenging programme to address the growing need for Child Welfare.

CIF broadly classified the programme to encompass following categories: -

- Abuse and Violence
- Trafficking
- Child Labour
- Conflict with Law
- Child Marriage
- Child Sexual Abuse
- Without Parental Care
- Street Children (Shelter)
- Birth Registration
- Armed Conflicts
- Disability
- Health issues?
- Education?
- Drug Abuse
- Gender bias (Girl Child)
- HIV - AIDS
- Missing Children

The 'ChildNet' System developed by Tata Consultancy Services Limited (TCS) assists to address the above mentioned categories. It also supports the following charters of United Nations:

- Child rights are specialized human rights that apply to all human beings below the age of 18
- Child rights defined by the United Nations and United Nations Convention on the Rights of the Child (UNCRC).
- According to the Integrated Child Protection Scheme (ICPS) Child Protection is about keeping children safe from a risk or perceived risk to their lives or childhood. It is about recognizing that children are vulnerable and hence reducing their vulnerability by protecting them from harm and harmful situations

<u>Child Tracking and Management Process followed by CIF</u>: -

1. Child / concerned Adult dials 1098
2. Call is connected to nearest Childline Contact Center / Collaborated agency
3. Information is collected in the ChildNet system.
4. Information is passed on to nearest CIF team or affiliated partners for intervention
5. Chidline team rushes to child without 60 minutes.
6. Appropriate Intervention and rehabilitation is done for each child by CIF team or associated Partners/NGOs.
7. Constant tracking is done to ascertain progress made for each Child

TCS provided comprehensive solution which would leverage Information Technology to capture and monitor progress of Child in Distress. In addition to this, TCS has advocated the use of distributed computing to address challenge of internet connectively in rural areas. Also TCS has leveraged open source technology to reduce the cost implication due to licensing.

TCS has developed a comprehensive system, to provide facts and figures which translate into visibility and accountability to Childline Childcare programme activities carried out in partnership with the District Women and Child Development (DWCD), State of Maharashtra, India.

III. METHODOLOGY

Tata Consultancy Services has employed open source techniques to facilitate a comprehensive solution to address the needs of CIF's Child Tracking System (ChildNet).

There are 5 phases for CIF process:

- **Feasibility Study:** In this phase TCS identified the overall objectives, scope of the project and outlined the desired outcomes. During this phase it was apparent that it was difficult to track the beneficiaries. It would be useful for CIF to get statistical data for trend analysis.
- **Requirements Elicitation:** There were two distinct topics which needed deliberation.

  1. *Existing Manual Process* – to obtain an outline of how activities were facilitated.

  2. *New System requirement* – to obtain a detailed insight on the new model of Childline Contact Center (CCC) requirements and map the need for change in process (as required).
- **Rapid Prototyping:** We developed a ChildNet System prototype to showcase the requirements thru all the envisaged Screens in the System with the detailed fields in each of the screens and the screens navigational workflow. Rapid prototyping is a unique way of gaining confidence and acceptance on the requirements, rather than the use of elaborate design documents which are technical in nature.
- **Preparation for Deployment:** This phase ran parallel to the application development phase, the rationale being to prepare CIF staff to accept the new system. TCS assisted in outlining the various call categories to effectively translate into call survey questionnaire in new system. In addition, TCS assisted in the formulation of the new process for effectively reporting statistics relating to achievements.
- **Deployment and Support:** In this phase there was a need to train and hand-hold the users while inducting them to the new philosophy. Mock runs were held to demonstrate how reporting would flow from both within and outside the system.

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IV. BENEFITS AND STATISTICS

The highlight of this innovation was CIF was able to showcase single truth of the programme and analyse trends which would help them in effectively improving and channeling efforts for the betterment of Children in Distress.

ChildNet application significantly classifies call records received at CCC which assisted in analysis various trends exhibited. Also the application assisted in tracking each Child for constant follow-up. The data from all CCC was aggregated and provided valuable insight into national statistics which would help identify the core challenge. ChildNet made it possible to study the patterns of calls, trends in child protection, hot spots of abuse and exploitation in the country, the quality and timelines of response and action by various stakeholders of child care and protection

ChildNet application assisted in formulation of the following reports:-

- **Monthly reports: -** Coherent account of activities conducted by various cities
- **Aggregated reports:** - Aggregation of all child intervention at Head office.
- **Call classification:** - Classifying various call interventions

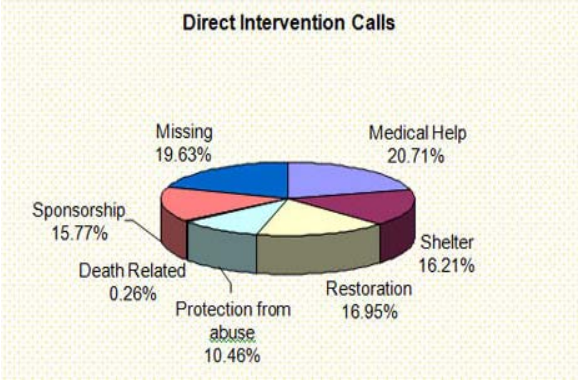

Figure 1. Direct Intervention Calls

Direct intervention calls were about missing children (-19.63%), medical (20.71%) and shelter issues (16.21%). Calls for repatriation (16.95%) were from children who had left home and now wanted to return. 10.46% of the calls were related to abuse. In response to such calls, the CIF teams met the caller/child to assess the situation and the intervention was planned taking the child's wishes into consideration. Calls for sponsorship (15.77%) included the need for access to education, health and foster care services. Calls for missing children were either from parents who reported that their child was missing, from a child who had been separated from his/her family or from a concerned adult or a CIF team member who had found the lost child.

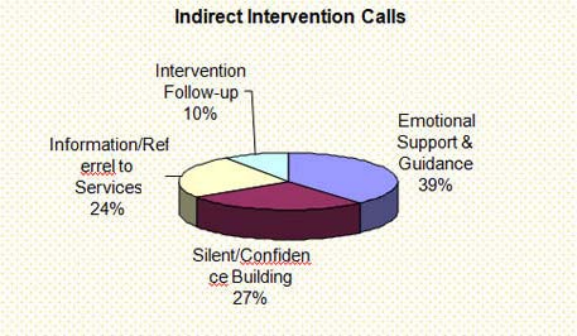

Figure 2. Indirect Intervention Calls

The highest number of indirect intervention calls was for emotional support and guidance (39%). Calls in this category come mainly from the child himself / herself when the child is emotionally upset because of family problems, peer group pressure, love relationships, school problems etc.

The ChildNet system provided detailed insight about all the various types of interventions where CIF has been involved.

TABLE I. All India Call Statistics 2009-10

No. of Cities	16	28	22	17	83
	North	South	East	West	Total
Medical Help	1,909	1,285	4,998	1,007	**9,199**
Shelter	972	3,007	2,165	1,056	**7,200**
Restoration	1,645	3,573	1,696	614	**7,528**
Protection from Abuse	869	1,834	865	1,077	**4,645**
Death Related	23	32	42	20	**117**
Sponsorship	813	876	3,411	1,905	**7,005**
Missing	2,438	2,059	3,044	1,179	**8,720**
Emotional Support & Guidance	155,022	103,181	112,003	169,225	**539,431**
Silent/Confidence Building	74,300	112,421	55,796	137,107	**379,624**
Information/Referral to Services	49,591	144,261	79,671	68,679	**342,202**
Intervention Follow-up	25,451	52,736	31,934	26,560	**136,681**
Unable to Locate Caller	706	1,885	1,311	766	**4,668**
Intervention Calls - I	**313,739**	**427,150**	**296,936**	**409,195**	**1,447,020**
Non Intervention Calls - II	**190,592**	**188,118**	**127,489**	**184,498**	**690,697**
Total I & II	**504,331**	**615,268**	**424,425**	**593,693**	**2,137,717**

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In conclusion ChildNet system was instrumental in articulation of the problem statement with strong statistical evidence. This ChildNet system also assisted CIF in visualizing, anticipating and streamlining processes to assist in addressing Vulnerable Children. This was an important factor to increase both accountability towards efforts by CIF staff by building the correct checks and balances to ensure data accuracy. The capture of the entire journey of the child once reported was unique and was a key contributor towards ensuring quality of data accuracy. The trend analysis on the various interventions churned out by the ChildNet system was viewed as a strong reference to enhance Child care policies in India by the Government of India. The trend analysis would highlight areas where apt focus is needed viz geography analysis, social behavior, type of intervention, socio economic conditions, child care services etc. The statistics were testimony to the fact that there is a huge deficit in basic Child care services and how basic intervention can assist a vulnerable child. CIF works closely with the local and central Government in order to strengthen the network and share best practices.

V. REFERENCES

Following are some of the reports referenced to compile statistics provided here: -

- ✓ Volume 6, Year 2010 – ChildNet Annual Report
- ✓ Volume 5, Year 2009 – CIF Annual Report
- ✓ Year 2008 – Annual Report

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